

CHINGGIS KHAAN INTL, Mongolia

WMO: 442910

Lat: 47.843N

Lon: 106.767E

Elev: 4364

StdP: 12.52

Time Zone: 8.00 (E08)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.2	-30.8	-40.6	0.6	-34.0	-37.1	0.8	-30.7	18.6	1.5	16.4	-2.5	2.3	180	0.713

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	86.8	59.5	83.1	58.4	79.8	57.5	63.9	78.2	62.2	75.6	60.8	73.7	11.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	88.9	67.9	57.7	83.5	66.9	56.0	78.5	65.6	31.5	78.0	30.2	75.3	29.1	73.7	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.2	21.1	18.4	DB	-38.8	93.0	3.6	5.5	-41.4	97.0	-43.4	100.2	-45.4	103.2	-48.0	107.2	(4)
(5)			WB	-38.5	66.8	4.0	2.1	-41.3	68.3	-43.7	69.5	-45.9	70.6	-48.8	72.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	-12.8	-4.8	18.2	36.4	48.3	60.7	65.4	60.8	49.2	31.7	7.8	-7.6	(6)	
	DBStd	28.88	8.68	10.15	12.66	9.50	9.43	7.75	6.00	6.98	8.43	8.51	12.54	8.78	(7)	
	HDD50	8775	1947	1534	986	416	148	7	0	4	114	568	1265	1786	(8)	
	HDD65	13076	2412	1954	1451	859	524	171	64	166	477	1032	1715	2251	(9)	
	CDD50	1336	0	0	0	7	94	328	478	340	89	0	0	0	(10)	
	CDD65	163	0	0	0	0	5	42	77	37	2	0	0	0	(11)	
	CDH74	2465	0	0	0	2	181	698	966	541	77	0	0	0	(12)	
	CDH80	714	0	0	0	0	41	206	298	158	11	0	0	0	(13)	
	WSAvg	6.8	3.6	4.7	6.8	9.1	10.4	8.9	7.8	7.2	7.4	6.5	4.7	3.8	(14)	
	PrecAvg	9.8	0.1	0.1	0.1	0.3	0.6	2.0	2.8	2.4	0.9	0.3	0.1	0.1	(15)	
(16) Precipitation	PrecMax	15.7	0.2	0.2	0.3	1.3	1.6	9.3	7.8	5.3	2.6	1.0	0.5	0.4	(16)	
	PrecMin	2.3	0.0	0.0	0.0	0.0	0.0	0.3	0.9	0.2	0.0	0.0	0.0	0.0	(17)	
	PrecStd	3.5	0.0	0.1	0.1	0.4	0.5	1.7	1.6	1.5	0.7	0.2	0.1	0.1	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	31.1	55.7	72.6	86.1	91.5	92.1	90.1	81.1	64.2	43.7	19.1	(19)	
		MCWB	13.9	23.5	39.0	46.9	54.5	60.0	62.2	61.7	55.2	44.3	33.4	16.4	(20)	
	2%	DB	11.6	24.7	48.8	66.2	79.3	85.5	87.0	84.2	75.9	57.7	39.0	14.0	(21)	
		MCWB	9.0	19.8	35.2	44.6	51.1	57.4	61.6	58.7	54.0	42.1	31.1	11.8	(22)	
	5%	DB	6.3	19.1	43.6	61.0	74.3	81.5	82.9	80.2	71.1	52.6	32.9	10.5	(23)	
		MCWB	4.6	15.7	32.5	42.0	49.2	56.2	60.7	58.9	51.9	39.2	27.1	8.8	(24)	
	10%	DB	2.0	13.8	38.2	56.2	69.3	77.8	79.3	75.8	67.0	48.0	28.2	6.8	(25)	
		MCWB	0.7	11.3	29.8	39.8	47.6	55.2	59.8	58.3	50.5	37.1	24.0	5.4	(26)	
	0.4%	WB	14.0	24.0	39.7	48.3	54.4	63.3	67.0	65.0	58.0	45.2	34.0	16.6	(27)	
		MCDB	17.3	31.0	54.2	69.3	82.7	79.8	79.6	81.0	73.9	61.3	42.5	19.2	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	9.0	19.8	35.9	44.8	52.0	60.4	64.8	63.0	55.9	42.7	31.0	12.0	(29)	
		MCDB	11.5	24.5	47.7	64.3	75.0	76.8	79.2	76.2	70.8	56.8	38.0	13.8	(30)	
	5%	WB	4.7	15.6	33.0	42.7	50.2	58.8	63.2	61.2	54.1	40.0	27.4	8.8	(31)	
		MCDB	6.4	19.1	42.9	59.5	73.2	73.8	77.2	73.4	68.1	51.2	32.5	10.5	(32)	
	10%	WB	0.7	11.4	29.9	40.2	48.0	57.3	61.8	59.7	51.8	37.6	24.0	5.4	(33)	
		MCDB	2.0	13.9	38.3	55.2	67.9	71.9	75.1	71.4	64.8	47.2	28.1	6.7	(34)	
	5% DB	MDBR	23.6	28.4	28.0	27.8	28.8	24.4	22.4	23.4	27.6	25.9	23.7	21.5	(35)	
		MCDBR	26.5	32.3	32.4	38.1	38.4	31.1	28.7	31.4	34.9	34.5	27.8	24.1	(36)	
		MCWBR	24.3	27.9	20.2	20.6	17.6	11.3	10.0	11.4	15.7	20.3	20.5	22.1	(37)	
		MCWBR	26.3	32.0	30.4	34.8	34.4	23.0	23.4	23.0	29.3	31.6	26.9	24.1	(38)	
(42) Clear-Sky Solar Irradiance	taub		0.240	0.263	0.303	0.349	0.380	0.390	0.375	0.356	0.311	0.283	0.258	0.242	(40)	
		taud	2.337	2.293	2.269	2.236	2.191	2.265	2.358	2.388	2.454	2.470	2.395	2.330	(41)	
	Ebn at Noon		273	288	291	285	279	277	279	281	286	278	263	256	(42)	
		Edh at Noon	22	29	35	41	44	42	37	35	29	24	21	19	(43)	
(44) All-Sky Solar Radiation	RadAvg		616	981	1433	1804	2078	2066	1947	1731	1455	1006	637	484	(44)	
	RadStd		35	60	59	75	145	115	109	113	79	51	35	31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page